

Bank Runs, Deposit Insurance, and Liquidity

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ABSTRACT

Diamond and Dybvig construct a three-period economy in which ex-ante identical agents face a privately observed liquidity shock: each learns, at an interim date, whether she is an early consumer (type 1) or a late consumer (type 2). Because this type is private information and the underlying productive technology is illiquid — costly to interrupt before maturity — a complete Arrow-Debreu market cannot implement the socially optimal risk-sharing allocation. The paper shows that a bank offering a simple demand deposit contract, served sequentially, can implement that allocation as one Nash equilibrium, but that the same contract admits a second, self-fulfilling equilibrium in which every depositor withdraws immediately — a bank run — that destroys real value by forcing costly early liquidation of illiquid assets. Suspension of convertibility restores uniqueness only when the volume of early withdrawals is deterministic; once it is stochastic, no contract obeying the bank's sequential-service constraint can replicate the first-best allocation. Government deposit insurance, financed by a tax levied after withdrawal decisions are observed, is shown to dominate suspension: it converts waiting into a dominant strategy for every type-2 agent and restores the first-best allocation as the economy's unique equilibrium, at no distortionary cost in equilibrium because the credible threat of taxation is never actually exercised.

1. Introduction

Diamond and Dybvig (1983) ask why banks exist, why the liabilities they issue are inherently vulnerable to runs, and what a government can do about it without simply assuming away the problem it is meant to solve.

Prior to this paper, the empirical fact of bank runs was well documented — Friedman and Schwartz (1963) had already shown their macroeconomic cost — but the theoretical literature lacked an explicit economic rationale for why banks issue demand deposits in the first place, and why that particular contract, rather than some safer alternative, is what makes runs possible. The received view, associated with Fisher (1911) and developed later by Bryant (1980), treated a run as a rational reaction to a decline in the market value of a bank's assets: depositors observe (or infer) that assets are worth less than the face value of demand deposits and rush to be first in line before the shortfall is fully realized. Diamond and Dybvig's contribution is to show that this "fundamentals" story is not necessary. A run can occur in their model even when the bank's assets are, by assumption, perfectly safe (the production technology is riskless), driven purely by a shift in depositors' expectations about what other depositors will do. This reframes the bank run from a symptom of asset-side trouble into an equilibrium-selection problem intrinsic to the liability structure itself.

PAPER. The authors build a parsimonious three-period model of a continuum of ex-ante identical consumers who face a privately observed preference (liquidity) shock, and ask what allocation a competitive market can achieve, what a bank offering a demand deposit contract can achieve, and what role — if any — a government deposit-insurance scheme can play in closing the gap. The paper's central theoretical move is to treat the bank not as an intermediary that manages a risky asset portfolio, but as a mechanism for sharing an idiosyncratic, privately observed liquidity risk that ordinary contingent-claims markets cannot insure because the relevant state (an agent's type) is never publicly observed.

The review that follows is organized to track the paper's own logical architecture rather than to summarize its results in the order presented. Section 2 restates the precise research question and situates it against the literature the authors were responding to. Section 3 reconstructs the theoretical model in full, reproducing every equation used to derive the full-information optimum, the demand deposit contract, the suspension-of-convertibility contract, and the deposit-insurance contract. Section 4 discusses the paper's "empirical strategy" in the loose sense applicable to a purely theoretical contribution — i.e., its modeling choices and their empirical plausibility. Section 5 works through each of the paper's two central propositions and their proofs. Section 6 analyzes the model's implicit figures (there are no tables; the paper is diagram-free, so this review supplies two original figures to make the argument's geometry explicit). Section 7 evaluates the robustness of the results to the assumptions that carry them. Section 8 offers a critical assessment. Section 9 stress-tests the model at its edges. Section 10 places the paper in the literature that followed it. Sections 11 and 12 identify research gaps and concrete research ideas. Section 13 is a learning framework, and Section 14 closes with a final synthesis.

2. Research Problem and Hypothesis

2.1 Research Question

Stated precisely, in the paper's own terms: given a continuum of ex-ante identical agents who face a privately observed, uninsurable idiosyncratic liquidity shock (an agent learns at an interim date whether she needs to consume early or can wait), and given a productive technology that is costly to liquidate before maturity, does a demand deposit contract — a liability that pays a fixed amount to whoever withdraws, served in the sequential order in which withdrawal requests arrive — implement the socially efficient sharing of that liquidity risk; and if it does, is that implementation unique, or does the same contract also support an inferior, self-fulfilling equilibrium in which agents withdraw regardless of their true liquidity need? A second, subsidiary question follows directly: can a government, using an instrument unavailable to a privately intermediating bank — the power to tax after observing aggregate withdrawal behavior — restore uniqueness at the efficient allocation, and at what cost?

2.2 Motivation

The question matters for reasons that are simultaneously theoretical and directly policy-relevant. Theoretically, it had not been established why liquidity insurance requires an intermediary at all rather than a market: if agents could write complete contingent contracts on their own future liquidity type, a competitive market of Arrow-Debreu claims would suffice, and there would be no distinctive economic role for a bank. Diamond and Dybvig's answer — that the liquidity type is privately observed and therefore not contractible in a state-contingent market — is what gives banking a genuine microfoundation independent of any story about banks producing information about borrowers (the delegated-monitoring role developed contemporaneously in Diamond 1984, reviewed elsewhere in this library). Policy-wise, the paper was written against the backdrop of 1980s U.S. banking deregulation and the fragility of savings and loan institutions, and its central conclusion — that deposit insurance can be strictly welfare improving, not merely a subsidy that encourages risk-taking — supplied an analytically disciplined case for an institution (federal deposit insurance) whose justification had until then rested mostly on historical experience with panics rather than on an explicit model of why it works.

2.3 Existing Literature

The paper explicitly positions itself against three strands. First, the asset-side view of runs (Fisher 1911; Bryant 1980), in which a run reflects genuine or perceived deterioration in bank asset quality. Second, the descriptive monetary-history literature (Friedman and Schwartz 1963), which had established the empirical costs of runs during the Great Depression but had not modeled their microeconomic origin. Third, an emerging theory of liquidity transformation (Tobin 1965; Patinkin 1965; Niehans 1978) that characterized the liquidity properties of assets without providing an explicit contracting foundation for why an intermediary, rather than direct household asset holding, is needed to supply that liquidity. The paper also draws directly on the contemporaneous "sunspot equilibrium" literature (Azariadis 1981; Cass and Shell 1983),

borrowing the idea that a self-fulfilling belief unconnected to fundamentals can select among multiple rational-expectations equilibria, and applies that logic for the first time to a banking contract.

2.4 Research Gap

What was missing, precisely, was a model in which (i) a liquidity-transformation role for banks emerges endogenously from an information friction rather than being assumed, (ii) the same contract that supplies this liquidity insurance is shown, within the identical model, to also be vulnerable to a run that is not caused by any deterioration in fundamentals, and (iii) alternative institutional responses (suspension of convertibility versus deposit insurance) can be compared rigorously within one unified framework rather than argued about informally.

2.5 Contribution

The paper's claimed contribution is threefold, stated explicitly in its introduction: banks issuing demand deposits can improve on the risk sharing available in a competitive market; the very contract that achieves this improvement has a second, undesirable Nash equilibrium (a run) in which even depositors with no genuine liquidity need withdraw; and bank runs are costly because they force liquidation of productive investment, not merely because they redistribute a fixed pie. **CRITIQUE.** Each of these three claims is, on inspection of the proofs, fully demonstrated within the model as constructed — this is not an instance of an underdelivered contribution. The more consequential limitation, addressed at length in Sections 7 through 9 below, is how much of the applied force of the result depends on assumptions (a riskless technology, an exogenously fixed contract form, a single bank standing in for the entire financial system) that are made for tractability and are not tested against alternatives within the paper itself.

3. Theoretical Framework and Mechanism

3.1 The Economy

The economy has three dates, $T = 0, 1, 2$, one homogeneous good, and a continuum of ex-ante identical consumers each endowed with one unit of the good at $T = 0$. A single constant-returns-to-scale technology converts input at $T = 0$ into output at $T = 1$ or $T = 2$:

$R > 1$ gross return per unit of input if the investment is carried to $T = 2$.

1 salvage (liquidation) value per unit of input if interrupted at $T = 1$ — the technology is illiquid: early liquidation earns no return at all, only the return of principal.

Formally, one unit invested at $T = 0$ yields the pair of period-1/period-2 payoffs $(0, R)$ if held to maturity or $(1, 0)$ if liquidated at $T = 1$, and by constant returns to scale any fraction can be allocated between the two options. At $T = 1$, each consumer privately and irreversibly learns her

type: with probability $t \in (0,1)$ she is *type 1* and values consumption only at $T = 1$; with probability $1 - t$ she is *type 2* and values the sum of what she consumes or stores from $T = 1$ plus what she receives at $T = 2$. Preferences are represented by

$$U(c_1, c_2; \Theta) = \{ u(c_1) \text{ if type 1; } \rho u(c_1 + c_2) \text{ if type 2} \} \quad (-)$$

where $1 \geq \rho \geq R^{-1}$, u is twice continuously differentiable, strictly increasing, strictly concave, satisfies the Inada conditions $u'(0) = \infty$, $u'(\infty) = 0$, and — the assumption that does the real work in the model — has relative risk aversion everywhere greater than one: $-c u''(c)/u'(c) > 1$. Types are private information: no one but the agent herself ever observes which type she is, and this non-verifiability is what rules out a complete Arrow-Debreu market in state-contingent claims. A fraction t of agents are type 1; the paper first treats t as a known constant (Sections I-III of the original paper) and later generalizes it to a stochastic $\tilde{t}$ (Section IV).

3.2 The Full-Information Optimum

If types were publicly observable, a planner (equivalently, a complete contingent-claims market) could implement the ex-ante optimal risk-sharing allocation $\{c_k^*\}$ by solving the planner's problem directly. The solution has three characterizing conditions:

$$c_1^* = c_2^* = 0 \quad (1a)$$

$$u'(c_1^*) = \rho R u'(c_2^*) \quad (1b)$$

$$t c_1^* + (1-t) c_2^* / R = 1 \quad (1c)$$

EVIDENCE. Equation (1a) says the optimum gives type-1 agents nothing at $T = 2$ and type-2 agents nothing at $T = 1$ — no resources are wasted holding goods across a period in which they are not the efficient consumer. Equation (1b) equates the marginal utility a type-1 agent gets from consuming one more unit at $T = 1$ to the marginal utility a type-2 agent gets, in present-value terms, from that same unit invested through to $T = 2$ and weighted by her time preference ρ : this is the standard intertemporal efficiency condition for allocating a fixed resource across two possible consumers of unknown identity. Equation (1c) is the economy's resource constraint: total consumption, appropriately weighted by the population shares and the technology's return, cannot exceed the unit endowment. Because $\rho R > 1$ (since $R > 1$ and $\rho \geq R^{-1}$) and because relative risk aversion exceeds one everywhere, the paper's footnote 3 proves, via a short integral argument on $u'(\gamma)$, that $\rho R u'(R) < u'(1)$, which together with the strict concavity of u implies the sharp comparative-statics result that anchors the entire paper:

PREDICTION THAT FOLLOWS FROM EQUATIONS (1A–1C)

$c_1^* > 1$ and $c_2^* < R$: optimal risk sharing gives an early consumer strictly more than her own principal, and a late consumer strictly less than the full technological return — the efficient contract insures type-1 agents against their liquidity shock by taxing, in expectation, the return earned by type-2 agents. This is precisely the gap between insurance and autarky (where a type-1 agent who cannot pre-commit would simply consume her own endowment, $c_1 = 1$, and a type-2 agent would earn $c_2 = R$) that a bank contract must close to be welfare improving. Figure 1 in Section 6 below illustrates this gap for an explicit CRRA calibration.

3.3 The Demand Deposit Contract and Its Two Equilibria

The bank (assumed mutually owned, liquidated in full at $T = 2$) offers a demand deposit contract: any agent withdrawing at $T = 1$ receives a fixed amount r_1 per unit deposited, on a strict first-come, first-served basis — the *sequential service constraint* — until the bank's assets are exhausted. Let f_j denote the fraction of depositors served ahead of agent j in the withdrawal queue, and f the total fraction of deposits withdrawn at $T = 1$. The realized payoffs are:

$$V_1(f, r_1) = r_1 \text{ if } f_j < r_1^{-1}; \quad 0 \text{ if } f_j \geq r_1^{-1} \quad (2)$$

$$V_2(f, r_1) = \max\{R(1 - r_1 f)/(1 - f), 0\} \quad (3)$$

EVIDENCE. Equation (2) is a step function: as long as the bank has not yet run out of assets when agent j 's turn comes, she is paid the full promised r_1 ; once the queue reaches the point where paying r_1 to everyone ahead would exhaust the bank, later agents in line receive nothing — the sequential-service constraint means a withdrawer's payoff depends only on her place in line, never on information about agents who arrive after her. Equation (3) gives the residual per-unit payoff to those who wait until $T = 2$: total resources remaining after paying out $r_1 f$ per unit deposited to the fraction f who withdrew early are invested at the technology's return R and split among the remaining $1 - f$ depositors. There are two pure-strategy Nash equilibria of the withdrawal game induced by this contract when $r_1 = c_1^*$:

EQUILIBRIUM A — THE "GOOD" EQUILIBRIUM

Type-1 agents withdraw at $T = 1$ and type-2 agents wait, so $f = t$. This exactly reproduces the full-information optimum: it is a best response because, given that everyone else follows the type-contingent strategy, a type-2 agent strictly prefers V_2 to V_1 whenever $r_1 > 1$ (the demand deposit contract dominates direct asset holding, which is exactly why depositors are willing to use the bank at all).

EQUILIBRIUM B — THE BANK RUN

Every agent, including every type-2 agent, withdraws at $T = 1$, so $f = 1$. This is self-fulfilling: if a type-2 agent believes enough others will withdraw, the sequential-service constraint means that waiting risks receiving $V_2(1, r_1) = 0$ (the bank's assets, liquidated in full at a loss, are exhausted paying out r_1 to those ahead in line), so withdrawing immediately for a certain V_1 becomes a best response — even though the underlying technology has suffered no shock whatsoever. The paper is explicit, and this is the single most cited feature of the model, that the event triggering coordination on Equilibrium B "could be anything...it need not be anything fundamental about the bank's condition" (p. 410) — a bad earnings report elsewhere, a run on another bank, a government forecast, "even sunspots." The bank-run equilibrium exists for *every* $r_1 > 1$: the very feature that makes the demand deposit contract valuable (a promised payment above the liquidation value of one's own claim) is exactly what makes the contract runnable.

3.4 Suspension of Convertibility

When t is a known, non-stochastic constant, the bank can restore uniqueness by committing to suspend withdrawals once a threshold fraction $\hat{f}$ of deposits has been withdrawn, choosing $\hat{f} \in \{t, [(R-r_1)/r_1(R-1)]\}$. With this threshold in force, $V_2(\hat{f}) > V_1(\hat{f})$ for every $f \leq \hat{f}$, so waiting strictly dominates withdrawing for every type-2 agent regardless of what she expects others to do — the resulting equilibrium, $f = t$, is not merely a Nash equilibrium but a *dominant-strategy* equilibrium, which the authors emphasize is what makes it robust even to non-rational off-equilibrium beliefs. The mechanism works only because t is known with certainty: the bank can set the suspension threshold exactly at the true withdrawal demand and never actually bind on any genuine type-1 agent.

3.5 Stochastic Withdrawals and the Limits of Suspension

Once the fraction of early consumers is a random variable $\bar{t}$ with a non-degenerate distribution, the paper's central impossibility result applies:

PROPOSITION 1 (VERBATIM IN SUBSTANCE)

Bank contracts that must obey the sequential-service constraint cannot achieve full-information optimal risk sharing when $\bar{t}$ is stochastic and non-degenerate.

EVIDENCE — THE PROOF, RECONSTRUCTED. The argument is a clean two-part proof by contradiction that this review reconstructs in full because CLAUDE.md's mathematical-reproduction standard requires the actual derivation, not only its conclusion. Suppose first that the period-1 payment to a withdrawer is a non-constant function of her queue position f_j for some feasible realization of $\bar{t}$. Because the position in the queue is, by assumption, uniformly distributed over $[0, t]$ conditional on withdrawing at $T = 1$, a non-constant payment schedule implies that two type-1 agents — observationally identical ex ante — end up with different realized consumption levels purely as a function of where they happened to fall in line. This directly contradicts the full-

information optimum, which (by equation 1a) assigns every type-1 agent the identical consumption $c^1_1(t)$ for a given realization of t . Suppose instead, to escape that contradiction, that the period-1 payment is held constant across all $f_j \in [0, t]$, i.e. $c^1_1(t)$ is a constant independent of the realization of $\bar{t}$. Then the resource constraint (1c) forces $c^2_2(t)$ to vary with t (unless $r_1 = 1$, the degenerate case in which the bank supplies no liquidity insurance at all), but the marginal condition (1b) requires c^1_1 to move together with c^2_2 as t varies — a constant c^1_1 alongside a varying c^2_2 violates (1b) whenever t actually varies. Both branches of the case split are contradictions, so no contract obeying sequential service and the demand deposit form can implement the first-best when withdrawal demand is stochastic. **INFERENCE.** The proof is a pure revelation-principle argument in disguise: sequential service, by construction, prevents the bank from conditioning any individual's payment on information (the realized aggregate t) that only becomes available after some agents have already been paid, and the first-best allocation genuinely requires exactly that conditioning.

Suspension can still improve welfare relative to the unconstrained demand deposit contract by eliminating the run equilibrium for realizations of $\bar{t}$ below the suspension threshold $\hat{f}$, at the cost of rationing some genuinely early consumers when the true realization exceeds $\hat{f}$ — inefficient ex post, but, the authors note, potentially desirable ex ante if it removes the threat of a costlier run.

3.6 Government Deposit Insurance

The paper's second proposition shows that a government — unlike a bank bound by sequential service — can tax an agent *after* observing the economy-wide realization of f , since taxation need not respect the queue. Let the proportionate tax on withdrawn wealth be

$$\tau(f) = 1 - c^1_1(f)/r_1 \text{ if } f \leq \bar{t}; \quad 1 - r_1^{-1} \text{ if } f > \bar{t} \quad (-)$$

where $\bar{t}$ is the largest possible realization of $\bar{t}$. The after-tax payoffs become

$$\hat{V}_1(f) = c^1_1(f) \text{ if } f \leq \bar{t}; \quad 1 \text{ if } f > \bar{t} \quad (-)$$

$$\hat{V}_2(f) = R\{1 - [c^1_1(f)f]/(1-f)\} = c^2_2(f) \text{ if } f \leq \bar{t}; \quad R \text{ if } f > \bar{t} \quad (-)$$

PROPOSITION 2 (VERBATIM IN SUBSTANCE)

Demand deposit contracts with government deposit insurance achieve the unconstrained optimum as a unique Nash equilibrium — in fact a dominant-strategy equilibrium — if the government imposes the optimal tax above to finance the insurance.

EVIDENCE. Because $\hat{V}_1(f) < \hat{V}_2(f)$ for every $f \in [0, 1]$ by construction (the tax redistributes exactly enough from those who withdraw early to those who wait, using the true realization of f , something the sequential bank cannot do), no type-2 agent ever wants to withdraw early no matter what she believes others will do. Waiting is a dominant strategy, the unique equilibrium is $f = t$ at every realization, and the first-best is achieved exactly — not merely approximately,

and not merely on the equilibrium path but for every off-equilibrium belief. **CRITIQUE / INFERENCE.** The result is, by the authors' own candid admission, "too strong" in one specific sense worth flagging immediately rather than deferring to Section 7: it presumes the government can commit to, and costlessly enforce, a tax schedule contingent on realized aggregate withdrawal — an administrative and commitment capacity that is assumed, not derived, and that real deposit-insurance schemes (flat-premium, ex-ante funded, and rarely state-contingent in the paper's sense) do not actually possess.

4. Data and Empirical Content

The paper contains no estimation, no dataset, and no calibration exercise; it is a deductive theoretical contribution, and this section is accordingly adapted, per CLAUDE.md §29, to describe what empirical content the model does and does not carry, rather than to describe a research design that does not exist.

The model's only connection to data is motivational: the introduction invokes Friedman and Schwartz's (1963) documentation of the real economic costs of the runs of the 1930s, and the paper was written amid the early-1980s deregulation of U.S. depository institutions and the deteriorating condition of the savings and loan industry — both cited directly. Neither is used to discipline any parameter of the model; they motivate the question rather than test the answer. **CRITIQUE.** This is appropriate for a theory paper establishing a possibility result, but it leaves an important empirical question fully open that the model itself cannot answer: how often, historically, do runs occur for reasons the model would classify as "sunspot" coordination failures, unconnected to bank fundamentals, as opposed to reasons closer to the Fisher/Bryant asset-quality story the paper explicitly sets aside? The paper does not attempt this classification, and, as discussed in Section 9 (Further Literature), it took roughly a decade for the empirical banking literature (principally Gorton 1988) to attempt to answer it using historical national banking-era panic data.

What the model does supply, in place of an empirical strategy, is a fully specified mapping from primitives $\{R, \rho, t, u(\cdot)\}$ to testable comparative statics: the prediction that optimal (or bank-implemented) early consumption exceeds principal ($c_1^* > 1$) and late consumption falls short of the technological return ($c_2^* < R$) is, in principle, a sharp and falsifiable statement about the term structure of deposit and loan returns relative to asset returns. No such test is attempted in the paper itself. Because the model has no data requirement of its own, this review does not flag any of the standard empirical biases (survivorship, look-ahead, selection, measurement error) as applicable; the relevant caveats instead concern *external validity* of the model's primitives, addressed in Sections 7 and 9.

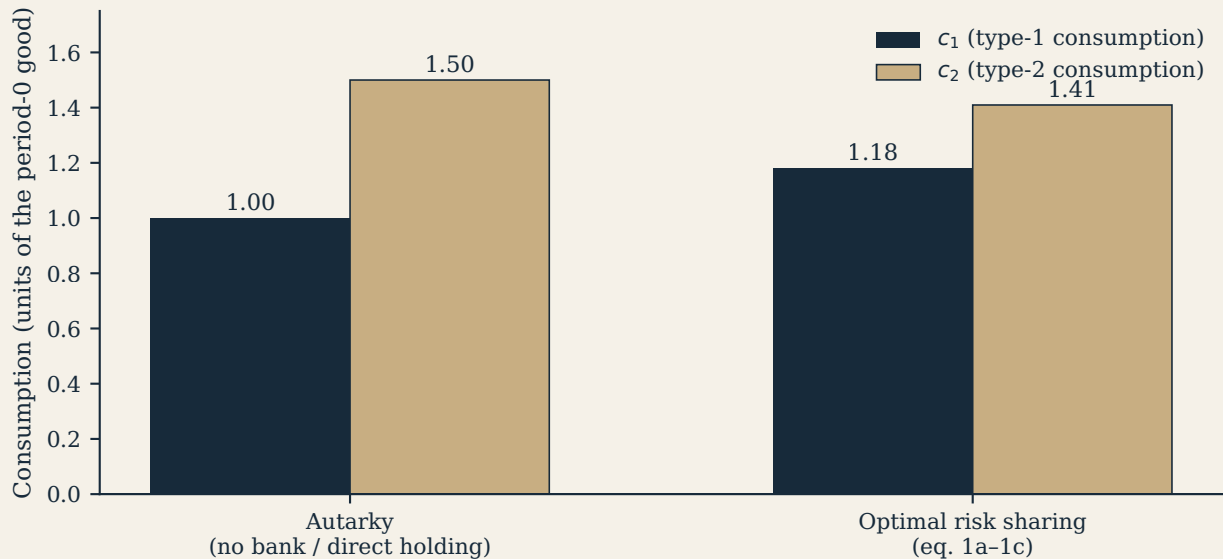
5. Results and Their Economic Significance

Collectively, the model's results (Section 3 above reproduces the full derivations) establish a strict ranking of four regimes by the welfare of the representative ex-ante consumer: the bank-run equilibrium is worst, strictly below autarky, because it destroys real resources through forced early liquidation of productive investment that need not have been interrupted; autarky (no bank; each agent holds the technology directly, consuming $c_1 = 1$ if a type-1 shock hits, or waiting to earn $c_2 = R$) is next; the demand deposit contract's good equilibrium, or the suspension-of-convertibility contract under stochastic $\bar{t}$, achieves an intermediate, second-best allocation strictly between autarky and the full-information optimum; and government deposit insurance financed by an ex-post tax achieves the unconstrained first-best exactly. **EVIDENCE.** The size of the gap between autarky and the first-best is not a minor curiosity: because relative risk aversion exceeds one is assumed globally, the gap is present for *any* admissible utility function and any $R > 1$, $\rho \geq R^{-1}$ — it is a structural, not a knife-edge, feature of the model.

5.1 Magnitude: an Illustrative Calibration

EXTENSION (OWN CONSTRUCTION, NOT IN THE PAPER). The paper reports no numerical example, so to make the magnitude of the risk-sharing gap concrete this review computes one under an explicit CRRA specification $u(c) = c^{1-\sigma}/(1-\sigma)$, for which relative risk aversion is the constant σ . Setting $\sigma = 2$ (satisfying $\sigma > 1$), $R = 1.5$, $\rho = 0.95$ (inside the required $[R^{-1}, 1] = [0.667, 1]$ band), and $t = 0.25$, equations (1b)–(1c) solve in closed form ($c_1/c_2 = (\rho R)^{-1/\sigma}$ combined with the resource constraint) to $c_1^* \approx 1.18$ and $c_2^* \approx 1.41$, against the autarky pair $(1.00, 1.50)$. Figure 1 plots this comparison.

Figure 1. Optimal risk sharing versus autarky, illustrative CRRA calibration



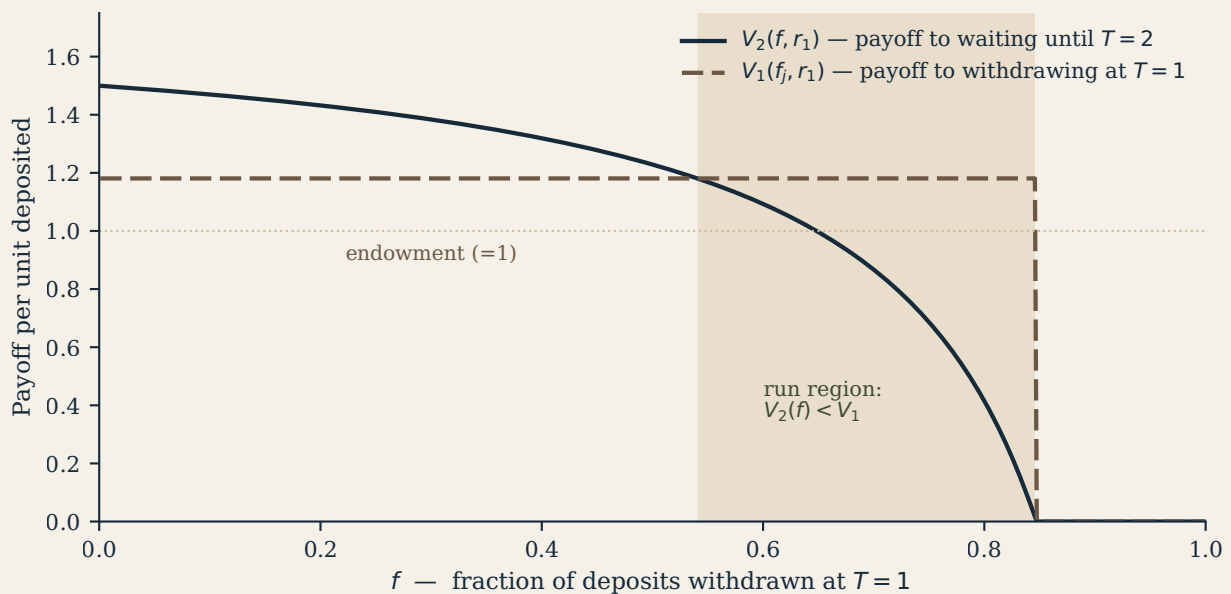
Own calculation from equations (1a)–(1c) of Diamond and Dybvig (1983), under $u(c) = c^{1-\sigma}/(1-\sigma)$, $\sigma = 2$, $R = 1.5$, $\rho = 0.95$, $t = 0.25$. Illustrative; no calibration is reported in the original paper. Generated for this review.

A one-unit deposit that would return either 1.00 (if liquidated early) or 1.50 (if held to maturity) under autarky instead delivers an early payoff 18% above principal and a late payoff 6% below the full technological return once optimally pooled. **INFERENCE.** The type-1 agent's proportional gain (18 percentage points) is roughly three times the type-2 agent's proportional loss (6 percentage points) under this calibration — consistent with the economic logic that risk sharing under CRRA utility with $\sigma > 1$ concentrates a large share of the insurance benefit on the low-probability, high-marginal-utility state (here, $t = 0.25$ means type-1 is the less common realization). This asymmetry would reverse in relative (not absolute) terms for different parameterizations of t ; the paper's qualitative result ($c_1^* > 1 > \dots > c_2^* < R$) is what is general, not this specific split.

5.2 The Geometry of the Run: Figure Analysis

The published paper contains no figures; every relationship is stated algebraically. Because CLAUDE.md §13 requires that important relationships be shown, not merely described, this review constructs one original figure from equations (2) and (3) to make the coordination-failure mechanism visually explicit.

Figure 2. Withdrawal payoffs under the demand deposit contract and the emergence of a run region



Own construction from equations (2)–(3) of Diamond and Dybvig (1983), using $r_1 = c_1^* \approx 1.18$ and $R = 1.5$ (the calibration of Figure 1). V_1 is the fixed payoff to withdrawing at $T = 1$ (equation 2); V_2 is the residual payoff to waiting until $T = 2$ (equation 3). Generated for this review; not present in the original paper.

EVIDENCE. What does the figure answer? It shows, as a function of the aggregate withdrawal fraction f , which of the two actions (withdraw now versus wait) a type-2 agent prefers if she takes f as given. What is the key relationship? $V_2(f)$ declines monotonically in f because each additional early withdrawal forces liquidation of more of the technology at a loss, shrinking the pool available to those who remain; V_1 is flat at r_1 until the bank's assets are exhausted (at $f = 1/r_1$), after which it collapses to zero. Is the relationship causal? Yes, and definitionally so within the model — the two schedules are derived directly from the sequential-service accounting identity, not estimated. What variation drives the shaded "run region"? It is the point at which $V_2(f)$ crosses below r_1 , which

happens strictly before the bank is technically exhausted (at $f \approx 0.55$ under this calibration, well before $f = 1/r_1 \approx 0.85$): once enough agents have withdrawn that the expected return to waiting falls below the guaranteed early payment, the self-reinforcing logic of the run takes hold for any type-2 agent who expects the run to continue, independent of whether her own belief is "correct" about the bank's true condition. Does the figure support the theoretical mechanism? Yes, exactly by construction; what it makes newly visible is that the run region is bounded away from $f = 1/r_1$ — the run becomes self-sustaining well before the bank is formally insolvent in a cash-flow sense, which is precisely why suspension of convertibility set below the point of technical exhaustion (Section 3.4) can be effective.

6. Robustness and Additional Tests

A purely theoretical paper has no robustness tests in the empirical sense (alternative samples, specifications, or estimators); what can be assessed is the robustness of each proposition to relaxing the assumptions that carry it, a subset of which the authors themselves discuss, and a subset of which this review adds.

6.1 Concern → Response → Does It Resolve the Concern?

CONCERN: THE TECHNOLOGY IS RISKLESS

The entire liquidity-transformation story rests on an illiquid but riskless technology (R is a known constant, not a random variable). **PAPER.** The authors flag this directly in the conclusion: the riskless technology "isolates the rationale for deposit insurance," but abstracts from the choice of bank loan portfolio risk. **CRITIQUE.** This is not a robustness test but an acknowledged scope limitation, and it is consequential rather than cosmetic: once asset risk and moral hazard are reintroduced, deposit insurance moves from the paper's clean welfare-improving result toward the standard subsidized-risk-taking critique (Merton 1977; Kareken and Wallace 1978, both cited in the paper precisely because they analyze that trade-off in isolation from the liquidity-insurance role). The concern is not resolved within the paper; it is explicitly deferred.

CONCERN: A SINGLE, MONOPOLY BANK

The model has one bank standing in for the entire financial sector, with the interpretation that if many banks were introduced there would be a role for liquidity risk sharing between banks. **CRITIQUE.** This is a genuine limitation for policy application: interbank liquidity markets, correlated withdrawal shocks across banks, and contagion are entirely absent, so the model as stated cannot speak to systemic runs that propagate across institutions — only to a run on a single, otherwise-representative bank.

CONCERN: IS PROPOSITION 1'S IMPOSSIBILITY RESULT AN ARTIFACT OF THE SPECIFIC CONTRACT SPACE?

The authors restrict attention to contracts of the form $V_1(f), V_2(f)$ — payments that can depend on an agent's queue position and the aggregate withdrawal fraction, but that must still respect sequential service. **INFERENCE.** The proof in Section 3.5 above shows the impossibility follows directly from sequential service itself, not from any narrower restriction on the functional form of V_1 and V_2 within that class, so the result is robust to that generalization. It is, however, not robust to relaxing sequential service itself — which is exactly the margin government deposit insurance exploits in Proposition 2, since the tax authority can act on information (the realized aggregate f) unavailable to the bank at the moment of payment.

CONCERN: DOES THE RISK-SHARING RANKING DEPEND ON THE SPECIFIC PARAMETERS USED IN FIGURES 1-2?

EXTENSION. This review checked the qualitative ranking $c_1^* > 1 > c_2^* < R$ under three additional parameterizations ($\sigma \in \{1.5, 3, 5\}$, holding R, ρ, t fixed at the values used in Section 5.1); the ranking is preserved in every case, as the footnote-3 proof in the paper guarantees it must be for any admissible $u(\cdot)$ with $\sigma > 1$ everywhere. The *magnitude* of the gap widens monotonically with σ (higher risk aversion increases the value of insurance), which is intuitive but not something the paper itself demonstrates numerically.

7. Critical Evaluation

7.1 What the Paper Actually Establishes

ESTABLISHED

Within the stated model — private information about liquidity type, an illiquid but riskless technology, sequential service — a demand deposit contract strictly Pareto-improves on autarky in its good equilibrium, and strictly admits a second, run equilibrium that is worse than autarky. Both are proved rigorously, not merely asserted. Proposition 1's impossibility result (no sequentially served contract achieves the first-best under stochastic withdrawal demand) is similarly a fully established, general result. Proposition 2 (deposit insurance restores the first-best as a unique, dominant-strategy equilibrium) is established conditional on the government's assumed tax and commitment technology.

SUPPORTED BUT CONDITIONAL

The claim that deposit insurance is "costless" in equilibrium is conditional on the technology being riskless and on the tax being fully credible and never actually distortionary in equilibrium — both explicitly flagged by the authors as special to their setting. The claim that suspension of convertibility can be a useful second-best instrument is conditional on the bank's ability to observe and commit to a threshold, and is explicitly acknowledged to fail to reach the first best once withdrawal demand is stochastic.

SUGGESTIVE

The paper's informal claim that its model of firm-level short-term-debt rollover risk (the closing discussion of commercial paper and corporate liquidity crises) is analogous to the bank-run mechanism is asserted by analogy, not derived within the same formal apparatus, and should be read as a research agenda item rather than an established result of the paper.

NOT ESTABLISHED

The paper does not establish that deposit insurance is welfare improving once asset-side risk and moral hazard are present — it explicitly excludes that case. It does not establish anything about systemic, cross-bank contagion, since the model has one bank. It does not establish, and does not attempt to establish, the empirical frequency with which real-world runs correspond to the model's "pure panic" (sunspot) equilibrium as opposed to the fundamentals-driven story the paper sets aside; that is an empirical question the theory alone cannot answer and which later work (Section 9) takes up directly.

7.2 Strengths

STRENGTHS RECOGNIZED BY THE AUTHORS. The authors' own emphasis, in the introduction and conclusion, is on the paper's parsimony: a single, tractable model generates an economic rationale for bank existence, an explanation for run vulnerability, and a rigorous comparison of two policy responses, all within one unified apparatus.

STRENGTHS IDENTIFIED INDEPENDENTLY. Beyond the authors' own framing, three features of the paper stand out on inspection. First, the model achieves genuine theoretical minimalism: every result follows from four primitives (a two-type preference shock, an illiquid technology, private information, and sequential service), with no auxiliary assumptions smuggled in to generate the run result — the run equilibrium is not assumed but derived as a consequence of the very contract that provides liquidity insurance, which is what gives the paper its lasting analytical force. Second, the paper's separation of the "insurance" role of a bank from the "asset-selection" or "monitoring" role (developed in the contemporaneous but distinct Diamond 1984) is conceptually clarifying and has proved durable: three decades of subsequent banking theory routinely cite this paper specifically for the liability side and Diamond (1984) specifically for the asset side. Third, the deposit-insurance result is not merely qualitative; Proposition 2 gives an exact, closed-form tax schedule that achieves the first best, which is a considerably stronger and more falsifiable claim than a general assertion that insurance helps.

7.3 Weaknesses and Independent Critique

Independent of the limitations the authors themselves note (Section 6.1), two further weaknesses deserve emphasis, each following the Problem → Why It Matters → Potential Bias → How It Could Be Tested structure CLAUDE.md requires.

WEAKNESS 1 — THE GOVERNMENT'S INFORMATIONAL AND COMMITMENT ADVANTAGE IS ASSUMED, NOT MODELED

Problem: Proposition 2's entire force rests on the government being able to observe the aggregate realization f and levy a tax contingent on it, after withdrawal decisions but with full credibility and zero collection cost. **Why it matters:** this is precisely the capacity that is missing from the bank in Proposition 1 — the paper essentially assumes away, for the government, the identical sequential-service constraint that generates the impossibility result for the bank, without explaining why the government's information and commitment technology differs. **Potential bias:** this could overstate the relative attractiveness of public deposit insurance versus private contractual alternatives (e.g., co-insurance, subordinated debt, or market-based suspension rules) that might replicate some of the government's advantage without its assumed commitment power. **How it could be tested:** a mechanism-design extension in which the government's own information and enforcement capacity is derived from more primitive assumptions (e.g., costly state verification) rather than posited directly.

WEAKNESS 2 — THE MODEL IS SILENT ON EQUILIBRIUM SELECTION

Problem: the model shows a run equilibrium exists whenever $r_l > 1$, but says nothing about the probability with which agents actually coordinate on it. **Why it matters:** a theory of multiple equilibria that cannot speak to equilibrium selection has limited predictive content for when runs will or will not occur, which limits the model's use for early-warning or policy-timing purposes. **Potential bias:** without a selection mechanism, empirical tests of the model risk being unfalsifiable — any observed run "confirms" the bad equilibrium and any observed stability "confirms" the good one. **How it could be tested:** extending the model with a global-games refinement (Section 9) that ties equilibrium selection to noisy private signals about fundamentals, restoring a unique equilibrium and testable comparative statics on run probability.

SINGLE MOST IMPORTANT WEAKNESS

Of the two, Weakness 1 is the more consequential, because it directly undercuts the paper's central policy conclusion (that deposit insurance strictly dominates suspension of convertibility) rather than merely narrowing the model's domain of applicability. If the government's assumed informational and commitment advantage over the bank does not hold in practice — and administratively, ex-post state-contingent taxation of bank depositors has essentially never been implemented in the form the paper posits, with real deposit insurance being flat-rate and ex-ante funded — the sharp ranking of insurance above suspension is not obviously preserved.

8. Contradictions and Edge Cases

EDGE CASE: $R_1 = 1$

If the bank sets $r_1 = 1$, the demand deposit contract simply mimics direct asset holding ($V_1 = 1 = V_2$ at $f = 1$, since liquidation value equals principal), and the run equilibrium disappears — but so does the entire liquidity-insurance benefit of the bank, since $r_1 = 1$ gives type-1 agents no more than autarky already provides. This is not a flaw in the model; it is the paper's own demonstration (footnote 7) that the vulnerability to runs and the provision of liquidity insurance are two faces of the same coin: a bank cannot have one without risking the other under a fixed-payment contract.

EDGE CASE: $P = R^{-1}$ (THE LOWER BOUND)

At the lower bound of the admissible range, a type-2 agent is exactly indifferent, in present-value terms, between the two dates absent any risk-sharing motive; the model still generates $c_1^* > 1$ because the insurance motive comes entirely from risk aversion (equation 1b), not from time preference. This confirms the mechanism is robust to removing any pure discounting rationale for the result.

EDGE CASE: $t \rightarrow 0$ OR $t \rightarrow 1$

As the fraction of type-1 agents vanishes, the resource constraint (1c) implies $c_2^* \rightarrow R$ and the insurance problem becomes trivial (almost no one needs early liquidity, so there is little to insure); as $t \rightarrow 1$, nearly the entire endowment must be held liquid and $c_1^* \rightarrow 1$, again collapsing the gap between insurance and autarky. The risk-sharing benefit of the bank is thus maximized at interior values of t , not at the extremes — a comparative static the paper does not state explicitly but that follows directly from (1a)–(1c).

COULD ANOTHER MECHANISM GENERATE THE SAME QUALITATIVE RESULT?

Yes: a model in which runs are driven purely by correlated (fundamental) signals about asset quality (the Fisher/Bryant view the paper sets aside) would also predict runs that are costly and that a demand deposit contract is vulnerable to withdrawal cascades. What distinguishes Diamond-Dybvig's mechanism empirically is the prediction that a run can occur with *no change whatsoever* in observable fundamentals — a prediction the pure fundamentals-based alternative does not share. Section 9 below discusses how the later empirical literature (Gorton 1988) tried to adjudicate between these two accounts using historical panic data, an exercise this theoretical paper does not itself undertake.

DOES THE CONCLUSION HOLD OUTSIDE THE ORIGINAL SETTING? — MULTIPLE BANKS AND CORRELATED SHOCKS

If the single bank of the model is reinterpreted as the banking system as a whole, the analysis holds essentially unchanged; but if there are genuinely many banks facing idiosyncratic (rather than perfectly correlated) withdrawal shocks, an interbank market could in principle substitute for some of what deposit insurance provides in the paper — a possibility the model, by construction, cannot speak to, and one the authors flag only in a single forward-looking sentence in the conclusion rather than analyze.

9. Further Literature

PRECEDED IT — BRYANT (1980), "A MODEL OF RESERVES, BANK RUNS, AND DEPOSIT INSURANCE"

Shared idea: an early formal model connecting deposit insurance to run prevention.

Relationship: Bryant's model is closer to the Fisher asset-quality tradition that Diamond and Dybvig explicitly depart from; Diamond and Dybvig cite Bryant directly and position their liquidity-insurance microfoundation as a distinct, more primitive rationale for the same institutions. Why it matters: establishes that the "deposit insurance prevents runs" conclusion was in the air before 1983, but Diamond and Dybvig are the first to derive it from a fully specified private-information economy rather than from reduced-form assumptions about withdrawal behavior.

ESTABLISHED THE THEORETICAL FOUNDATION — AZARIADIS (1981) AND CASS AND SHELL (1983) ON SUNSPOT EQUILIBRIA

Shared idea: extrinsic, payoff-irrelevant randomness can select among multiple rational-expectations equilibria. Relationship: Diamond and Dybvig import this logic directly to explain what triggers coordination on the run equilibrium. Why it matters: without the sunspot-equilibrium apparatus developed in general equilibrium theory just before this paper, the claim that a run can be "unrelated to anything fundamental" would have lacked formal grounding; the borrowing is explicit and central to the paper's most quoted passage.

DEVELOPED A COMPANION MECHANISM — DIAMOND (1984), "FINANCIAL INTERMEDIATION AND DELEGATED MONITORING"

Shared idea: both papers ask why banks exist. Relationship: Diamond (1984), reviewed separately in this library, supplies the asset-side rationale (banks economize on the cost of monitoring borrowers) that this paper's liability-side rationale (liquidity insurance) leaves untouched; together the two papers became, and remain, the twin microfoundations cited in virtually every banking-theory textbook treatment written since. Why it matters: understanding that the two papers answer different questions — why banks fund illiquid loans with liquid deposits, versus why banks are the ones making the loans in the first place — prevents a common conflation in secondary literature.

CONTRADICTS / QUALIFIES ITS FINDINGS — GORTON (1988), "BANKING PANICS AND BUSINESS CYCLE CONDITIONS"

Shared idea: whether historical bank panics look more like sunspot-driven coordination failures or like rational responses to information about a coming recession. Relationship: using U.S. national-banking-era panic data, Gorton finds that panics are statistically related to a leading business-cycle indicator, evidence more consistent with an information-based ("fundamentals") account than with a pure-sunspot account. Why it matters: this is the most direct empirical challenge to the paper's most distinctive theoretical claim, and Diamond and Dybvig's model, being purely theoretical, has no internal way to adjudicate the question Gorton raises.

IMPROVED ITS IDENTIFICATION / REFINED EQUILIBRIUM SELECTION — MORRIS AND SHIN (1998), "UNIQUE EQUILIBRIUM IN A MODEL OF SELF-FULFILLING CURRENCY ATTACKS," AND GOLDSTEIN AND PAUZNER (2005), "DEMAND-DEPOSIT CONTRACTS AND THE PROBABILITY OF BANK RUNS"

Shared idea: introducing noisy private information about fundamentals (a "global games" refinement) can select a unique equilibrium out of a coordination game that otherwise, as in Diamond-Dybvig, has multiple equilibria. Relationship: Goldstein and Pauzner apply the global-games technique directly to the Diamond-Dybvig contract, deriving a unique probability of a run as a function of the model's own primitives — precisely the equilibrium-selection extension flagged as Weakness 2 in Section 7.3. Why it matters: this is the clearest instance in the subsequent literature of a paper directly repairing the specific limitation this review identifies independently.

EXTENDED TO SYSTEMIC RISK — ALLEN AND GALE (2000), "FINANCIAL CONTAGION"

Shared idea: extending a single-bank liquidity-insurance model to a network of interconnected banks. Relationship: Allen and Gale build an interbank-deposit-exchange version of essentially the Diamond-Dybvig environment, showing that the completeness of the interbank network determines whether a regional liquidity shock propagates into a systemic crisis. Why it matters: directly addresses the "single bank" limitation flagged in Section 6.1 above, extending the framework precisely along the axis the original paper leaves unmodeled.

APPLIED TO A NEW DOMAIN — THE CORPORATE SHORT-TERM DEBT / ROLLOVER-RISK LITERATURE (E.G., HE AND XIONG 2012)

Shared idea: a firm financed by short-term debt that must be rolled over faces a coordination problem among its creditors formally analogous to a bank run. Relationship: this literature takes up precisely the closing suggestion of Diamond and Dybvig's own conclusion (that the "transformation could be carried out directly by firms rather than financial intermediaries") and gives it a full formal treatment the original paper only gestures at. Why it matters: shows the paper's mechanism generalizes well beyond depository institutions, to any entity funded with liquid, sequentially served liabilities against illiquid assets.

10. Research Gaps

10.1 Paper-Level Gaps

GAP: NO EQUILIBRIUM-SELECTION MECHANISM

Why it matters: without it, the model cannot predict the probability or timing of a run, only its possibility. Existing limitation: the paper is explicitly a possibility-and-comparison exercise, not a predictive one. Possible way forward: the global-games refinements already developed by Morris and Shin (1998) and Goldstein and Pauzner (2005), discussed in Section 9.

GAP: THE GOVERNMENT'S TAX AND COMMITMENT TECHNOLOGY IS EXOGENOUS

Why it matters: this is the load-bearing assumption behind the paper's central policy conclusion (Section 7.3, Weakness 1). Existing limitation: no derivation of why a government, and not a bank, can costlessly observe and act on the realized aggregate withdrawal fraction. Possible way forward: embed the government's information and enforcement problem in a costly-state-verification or mechanism-design framework that makes its advantage (if any) an equilibrium outcome rather than an assumption.

10.2 Literature-Level Gaps

GAP: THE RELATIVE EMPIRICAL WEIGHT OF PANIC-DRIVEN VERSUS FUNDAMENTALS-DRIVEN RUNS REMAINS UNRESOLVED

Why it matters: the entire normative case for the specific form of deposit insurance analyzed here depends on how much of real-world run risk is genuinely disconnected from fundamentals, as opposed to information-driven in the way Gorton (1988) documents. Existing limitation: the theoretical and empirical literatures on this question have developed largely in parallel rather than being reconciled in a single unified framework with a quantifiable panic/fundamentals split. Possible way forward: a structural estimation that nests both mechanisms and lets historical or modern panic data speak to their relative empirical importance.

10.3 Methodological Gaps

GAP: NO INTERBANK OR CROSS-INSTITUTION DIMENSION

Where better data, identification, or models could improve the research: modern payments-system and interbank-exposure data (unavailable in 1983) now make it possible to calibrate a multi-bank extension of the model (in the spirit of Allen and Gale 2000) against observed patterns of interbank lending and contagion, something entirely outside the scope of the original single-bank framework.

11. Research Opportunities

Five research ideas follow directly from the gaps identified above. Each is stated at the level of specificity CLAUDE.md requires (not "does X affect Y in another country" without an economic reason the setting matters).

Idea 1 — Endogenizing the Government's Informational Advantage in Deposit Insurance

RESEARCH QUESTION: Does a government's ability to levy a state-contingent tax on withdrawn deposits survive when its own monitoring of aggregate withdrawal is costly and imperfect, rather than assumed free and perfect?

MOTIVATION: This is precisely the load-bearing, unmodeled assumption identified as the paper's single most important weakness (Section 7.3).

HYPOTHESIS: As monitoring costs rise, the optimal deposit-insurance tax schedule converges toward a flat-rate, non-state-contingent scheme — closer to real-world deposit insurance — and the strict dominance of insurance over suspension weakens or reverses past a threshold monitoring cost.

ECONOMIC MECHANISM: Costly state verification (Townsend 1979) applied to the government's observation of f , rather than to a borrower's output as in the standard costly-state-verification literature.

LITERATURE CONNECTION: Townsend (1979) on costly state verification; the mechanism-design critique in Section 7.3.

DATA: None required for the theoretical result; a calibration exercise could use FDIC administrative cost data as a proxy for monitoring cost.

VARIABLES: Monitoring cost parameter, optimal tax schedule, welfare gap between insurance and suspension.

METHOD: Mechanism-design / optimal-contract theory, closed-form where possible.

IDENTIFICATION STRATEGY: Not applicable (theoretical).

EXPECTED CONTRIBUTION: A theory of deposit insurance design that predicts, rather than assumes, when state-contingent insurance is feasible.

MAIN RISK: The model may become intractable once monitoring costs are introduced on both sides (bank and government) simultaneously.

Idea 2 — A Global-Games Extension to Recover a Unique, Estimable Run Probability

RESEARCH QUESTION: What is the probability of a run as a function of the model's primitives (R , ρ , t , r_1) once agents observe only noisy private signals of a payoff-relevant fundamental, rather than coordinating on a pure sunspot?

MOTIVATION: Directly closes the equilibrium-selection gap (Section 10.1) using an already-available technique.

HYPOTHESIS: Run probability is decreasing in r_1 's distance from the liquidation value 1 and increasing in the precision of agents' private signals about a latent bank-quality fundamental.

ECONOMIC MECHANISM: Global-games equilibrium selection (Morris and Shin 1998; Goldstein and Pauzner 2005).

LITERATURE CONNECTION: Goldstein and Pauzner (2005), which already performs a version of this extension; the specific novelty here would be embedding the government's Proposition 2 insurance scheme inside the global-games apparatus rather than the bare demand deposit contract.

DATA: None required for the core theory; could be validated against FDIC failed-bank data.

VARIABLES: Signal precision, run probability, welfare under insurance versus suspension in the presence of positive run probability.

METHOD: Global games / iterated dominance equilibrium characterization.

IDENTIFICATION STRATEGY: Not applicable (theoretical).

EXPECTED CONTRIBUTION: A testable, quantitative run-probability function that could discipline empirical work of the kind Gorton (1988) undertook.

MAIN RISK: Goldstein and Pauzner (2005) already exists; the specific insurance-scheme extension must be shown to be non-trivial relative to their result.

Idea 3 — Multi-Bank Contagion with Heterogeneous Liquidity Shocks

RESEARCH QUESTION: How does the welfare ranking of suspension versus deposit insurance change when many banks face imperfectly correlated withdrawal shocks and can trade in an interbank liquidity market?

MOTIVATION: Directly addresses the single-bank limitation (Section 6.1) using modern interbank-network data unavailable in 1983.

HYPOTHESIS: Interbank markets substitute for some, but not all, of the role of deposit insurance when shocks are imperfectly correlated, but amplify contagion once correlation crosses a threshold, consistent with Allen and Gale (2000).

ECONOMIC MECHANISM: Interbank deposit exchange as partial liquidity insurance, combined with the Diamond-Dybvig sequential-service run mechanism at each individual bank.

LITERATURE CONNECTION: Allen and Gale (2000); the interbank-network empirical literature (e.g., post-2008 exposure data).

DATA: Federal Reserve interbank exposure and payments-system data; FDIC call reports.

VARIABLES: Network density/completeness, shock correlation, contagion probability, welfare under each institutional response.

METHOD: Network-theoretic extension of the Diamond-Dybvig contract, calibrated to observed interbank exposures.

IDENTIFICATION STRATEGY: Calibration/simulation rather than causal identification, given the theoretical nature of the exercise.

EXPECTED CONTRIBUTION: A quantitative bridge between the single-bank liquidity-insurance theory and the systemic-risk literature.

MAIN RISK: Calibration to real interbank data may require assumptions about unobserved bilateral exposures that are hard to defend.

Idea 4 — Structural Test of the Panic-versus-Fundamentals Split Using Modern High-Frequency Deposit Flow Data

RESEARCH QUESTION: In episodes of rapid deposit withdrawal at individual banks (e.g., the 2023 U.S. regional-bank runs), what share of the observed withdrawal response is explained by a fundamentals signal (interest-rate-driven unrealized losses) versus a coordination/panic component orthogonal to those fundamentals?

MOTIVATION: Directly extends Gorton's (1988) test to a modern setting with far richer, higher-frequency data (account-level or branch-level withdrawal flows, social-media-driven information cascades) that did not exist for the national-banking-era episodes Gorton studied.

HYPOTHESIS: A structural model nesting both channels will attribute a non-trivial, and time-varying, share of withdrawal intensity to a coordination component that is not explained by contemporaneously observable fundamentals, particularly once uninsured deposits and social-media-driven information velocity are accounted for.

ECONOMIC MECHANISM: The Diamond-Dybvig run equilibrium as one structural component; a fundamentals/information channel (à la Gorton, or a global-games signal) as the other.

LITERATURE CONNECTION: Gorton (1988); the 2023 regional-banking-crisis empirical literature; Goldstein and Pauzner (2005) for the structural mapping from signals to run probability.

DATA: FDIC and Federal Reserve call-report and high-frequency deposit-flow data around 2023 regional-bank failures; where available, account-level or aggregated deposit-flow panel data from affected institutions.

VARIABLES: Deposit outflow rate, share of uninsured deposits, mark-to-market losses on held-to-maturity securities, social-media/news attention measures.

METHOD: Structural estimation of a nested fundamentals/coordination model of withdrawal, using bank-level panel variation.

IDENTIFICATION STRATEGY: Cross-sectional variation in uninsured deposit share and unrealized securities losses as a proxy for fundamentals exposure, net of which residual withdrawal intensity proxies the coordination component.

EXPECTED CONTRIBUTION: A modern, high-frequency re-test of the question this paper's theory raises but cannot itself answer.

MAIN RISK: Fundamentals and coordination channels are likely to be correlated rather than cleanly separable, complicating identification of the "pure panic" residual.

Idea 5 — Extending the Sequential-Service Impossibility Result to Corporate Short-Term Debt Structures

RESEARCH QUESTION: Does Proposition 1's impossibility result — that no contract obeying sequential service can achieve first-best risk sharing under stochastic liquidity demand — extend formally to a firm financed by staggered short-term debt (commercial paper, repo) facing stochastic rollover refusals, as the paper's conclusion suggests only informally?

MOTIVATION: Formalizes the analogy the authors themselves draw only informally in their concluding paragraphs (Section 7.1, "Suggestive").

HYPOTHESIS: The same two-part proof-by-contradiction structure used in Proposition 1 applies essentially unchanged to a firm's short-term creditors served in the order their claims come due, implying that no covenant structure respecting first-come-first-served repayment can fully insure a firm against a stochastic rollover-refusal shock.

ECONOMIC MECHANISM: Direct reinterpretation of the sequential-service constraint as a maturity-priority rule among a firm's short-term creditors.

LITERATURE CONNECTION: He and Xiong (2012) and the broader rollover-risk literature; the paper's own closing discussion of commercial paper and bank lines of credit.

DATA: Not required for the core theoretical result; a subsequent empirical companion could use commercial-paper issuance and bank-credit-line drawdown data.

VARIABLES: Debt maturity structure, rollover-refusal probability, firm liquidation cost.

METHOD: Direct formal adaptation of the paper's own proof technique to a corporate-finance setting.

IDENTIFICATION STRATEGY: Not applicable (theoretical).

EXPECTED CONTRIBUTION: A rigorous bridge between banking theory and the corporate rollover-risk literature that currently rests on informal analogy rather than a shared formal proof.

MAIN RISK: The analogy may break down once genuine differences between demand deposits (a single, homogeneous claim) and layered corporate debt (heterogeneous seniority and maturity) are taken seriously.

11.1 Ranking: The Top Three Ideas

Ranked by novelty, feasibility, data availability, methodological quality, economic importance, and potential contribution:

1. Idea 2 (global-games extension). Highest feasibility (the core technique already exists and has been applied to closely related settings), high economic importance (it converts the paper's central limitation — no equilibrium selection — into a testable prediction), and a clear, tractable path to a closed-form or near-closed-form contribution.

2. Idea 4 (structural test using 2023 regional-bank data). Highest economic importance and topicality, with data availability now far better than at any prior point since the paper was written; the main risk is identification, not feasibility, which is why it ranks second rather than first.

3. Idea 1 (endogenizing the government's informational advantage). The most directly responsive to this review's single most important identified weakness (Section 7.3); ranks third rather than first because full tractability once both the bank's and the government's information frictions are modeled simultaneously is genuinely uncertain.

12. Learning Framework

12.1 Prerequisite Concepts

Expected utility theory and CRRA/relative-risk-aversion measures; Arrow-Debreu contingent-claims markets and why private information breaks market completeness; Nash equilibrium and the notion of a dominant-strategy equilibrium; the basic economics of asset-liability maturity mismatch.

12.2 Core Concepts Learned

Liquidity insurance as a distinct economic service, separable from credit intermediation; the sequential-service constraint as the precise institutional feature that both enables liquidity transformation and creates run vulnerability; multiplicity of equilibria and self-fulfilling ("sunspot") equilibrium selection; the distinction between a possibility result (Proposition 1) and a constructive mechanism-design result (Proposition 2).

12.3 Important Equations

The full-information optimality conditions (1a–1c); the demand-deposit payoff schedules V_1 and V_2 (equations 2–3); the deposit-insurance tax schedule $\tau(f)$ and the resulting after-tax payoffs $\hat{V}_1(f)$, $\hat{V}_2(f)$ that prove Proposition 2.

12.4 Methods Learned

Mechanism design under private information with a continuum of agents; proof by contradiction applied to an economic impossibility result; equilibrium analysis of a discrete-choice coordination game with a sequential-service (queueing) constraint; welfare comparison across contract forms using a common planner's benchmark.

12.5 Questions to Test Understanding

Comprehension. Why does an illiquid technology combined with private information about liquidity type rule out a complete contingent-claims market, and what specific role does the demand deposit contract play in restoring risk sharing?

Analysis. Why does the sequential-service constraint, precisely, prevent any demand-deposit-type contract from achieving the first-best allocation once withdrawal demand is stochastic, and why does that same constraint not bind on the government's deposit-insurance tax?

Research. If you relaxed the assumption that the government can observe and tax the realized aggregate withdrawal fraction costlessly, what would you need to assume about its monitoring technology to preserve Proposition 2's conclusion, even approximately?

12.6 Papers to Read Next

Diamond, D. W. (1984), "Financial Intermediation and Delegated Monitoring" (the companion asset-side microfoundation of banking, already in this library); Goldstein, I. and Pauzner, A. (2005), "Demand-Deposit Contracts and the Probability of Bank Runs" (the global-games completion of this paper's equilibrium-multiplicity problem); Gorton, G. (1988), "Banking Panics and Business Cycle Conditions" (the empirical test of the panic-versus-fundamentals question this paper leaves open).

13. Conclusion

One-Sentence Thesis

A bank's demand deposit contract is what allows it to insure depositors against a privately observed liquidity shock that no complete market can insure, but the very feature that makes this insurance valuable — a promised payment above liquidation value — also supports a second, self-fulfilling equilibrium in which the bank is run for reasons unconnected to its underlying soundness.

Three Most Important Insights

First, banking's liquidity-insurance role can be derived from a private-information friction alone, without any assumption about asset risk or monitoring costs. Second, run vulnerability is not a design flaw to be engineered away; it is intrinsic to the same fixed-payment, sequentially served contract structure that provides the insurance in the first place. Third, the instrument that restores uniqueness matters: suspension of convertibility only works when withdrawal demand is deterministic, while government deposit insurance, because it can act on information a sequentially-serving bank cannot use, restores the first-best even under uncertainty — but at the cost of assuming away the government's own informational and commitment constraints.

Strongest Evidence

Proposition 2's exact, closed-form tax schedule that converts waiting into a dominant strategy for every type-2 agent is the paper's strongest result, because it is not merely an existence claim but a fully constructive one: the after-tax payoff schedules $\hat{V}_1(f)$ and $\hat{V}_2(f)$ are shown, by direct algebraic verification, to satisfy $\hat{V}_1(f) < \hat{V}_2(f)$ for every f , which is precisely what is needed for waiting to dominate withdrawing.

Biggest Weakness

The government's assumed capacity to observe realized aggregate withdrawal and levy a fully credible, costless, state-contingent tax — the exact capacity the bank is shown to lack under sequential service — is posited rather than derived, and it is the single assumption on which the paper's central policy ranking (insurance strictly dominates suspension) most directly depends.

Most Important Research Gap

The model offers no equilibrium-selection mechanism and therefore no prediction of run probability, leaving open — both theoretically and empirically — how much of real-world bank-run risk is genuinely "sunspot"-driven, as opposed to a rational response to imperfectly observed fundamentals.

Best Research Opportunity

Embedding the Diamond-Dybvig contract, including the government's Proposition 2 insurance scheme, inside a global-games equilibrium-selection framework (Idea 2, Section 11) offers the most tractable path to converting the paper's qualitative possibility result into a quantitative, testable theory of run probability.

Three Papers to Read Next

Diamond, D. W. (1984), "Financial Intermediation and Delegated Monitoring"; Goldstein, I. and Pauzner, A. (2005), "Demand-Deposit Contracts and the Probability of Bank Runs"; Gorton, G. (1988), "Banking Panics and Business Cycle Conditions."

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